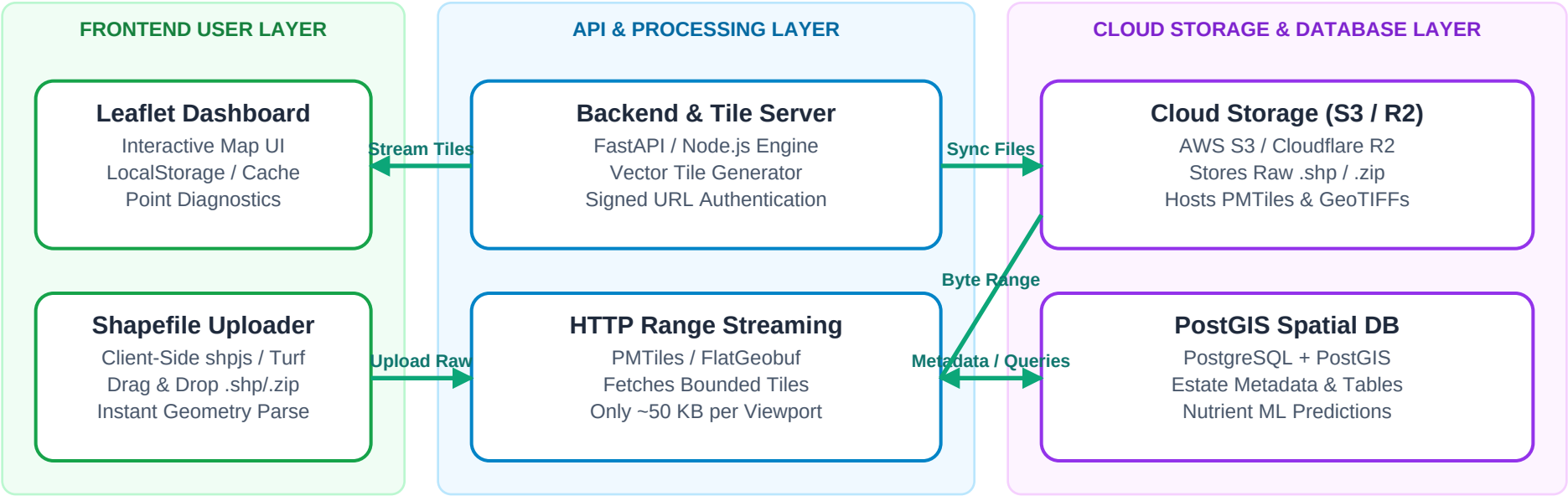


SmartPalm / PalmNex — Cloud GIS Architecture Framework

High-Scale Spatial Pipelines, Cloud Object Storage (S3/R2), PostGIS & Viewport Vector Streaming



System Component Specifications

Component Layer	Technology / Service	Primary Function & Data Handled	Storage & Performance
Client Dashboard	Leaflet.js + Turf.js + shpjs	Interactive visualization, shapefile parsing, point nutrient diagnostics, layer toggling.	IndexedDB / LocalStorage (~5-10 MB client cache).
Cloud Storage	AWS S3 / Cloudflare R2	Stores raw Shapefiles (.shp, .zip), PMTiles archives, and high-res GeoTIFF raster maps.	Scalable to Terabytes; HTTP Range Request streaming.
Spatial Database	PostgreSQL + PostGIS	Stores estate polygons, metadata catalog, soil nutrient levels, and fertilizer recommendations.	Millisecond spatial indexing (R-Tree / GIST indexes).

Vector Streaming	PMTiles / Mapbox MVT	Streams bounded spatial tile data corresponding only to the user's active map viewport.	Ultra-low bandwidth (~50 KB per viewport load).
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